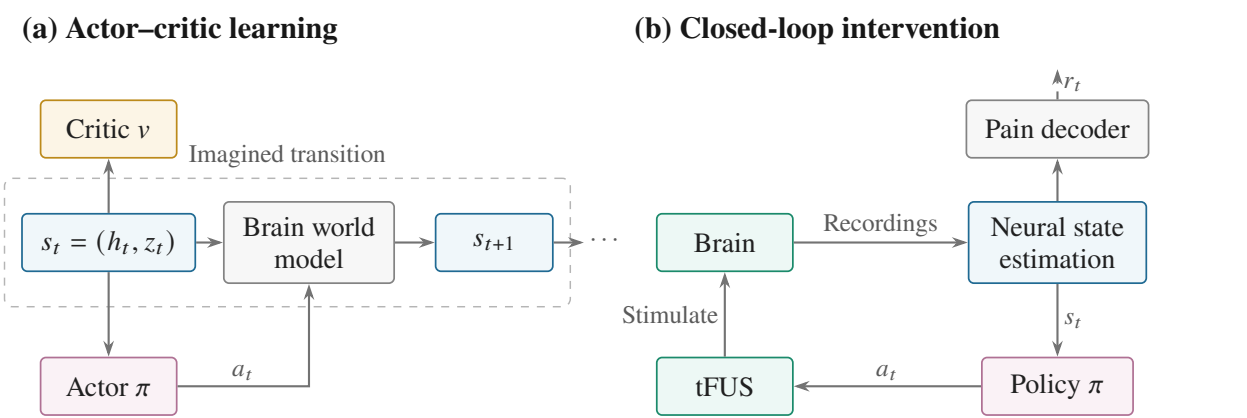

(a) Actor–critic learning
Critic v
Imagined transition
$s_t = (h_t, z_t)$
Brain world model
s_{t+1}
Actor π
a_t
(b) Closed-loop intervention
r_t
Pain decoder
Brain
Recordings
Neural state estimation
Stimulate
s_t
tFUS
a_t
Policy π